

Energy efficiency and Spectral efficiency tradeoff in semantic communication enhanced by Device to device communication

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Abstract—The explosive growth of connected mobile devices and data-heavy applications running on these devices is pushing wireless networks to their limits, posing a critical challenge: achieving high spectral efficiency (SE) while minimizing energy consumption. Conventional systems transmit every bit of raw data, even when much of it is irrelevant to the actual task, wasting precious spectrum, power, and time—problems that will only intensify in 6G networks. This paper proposes a effective solution by integrating semantic communication with device-to-device (D2D) links. Semantic communication extracts and transmits only the task-critical meaning of the data, dramatically reducing redundant bits and overall traffic. D2D enables nearby devices to communicate directly, shortening transmission distances, cutting path loss, latency, and energy use even further. Extensive simulations across a wide range of signal-to-noise ratios (SNR) reveal that both energy efficiency (EE) and SE improve markedly with increasing SNR, primarily because of fewer errors and retransmissions. However, beyond a SNR threshold, gains taper off and saturated, reflecting the classic EE-SE tradeoff. By intelligently combining semantic transmission and D2D, proposed approach achieve greener, more efficient networks without compromising reliability which makes this approach highly promising for sustainable 6G ecosystems where spectrum and energy will be severely constrained.

Index Terms—Semantic Communication, Device-to-Device (D2D) Communication, Spectral Efficiency, Energy Efficiency, Modulation Techniques, SNR, Wireless Performance, Path Loss, Compression, Cooperative Links

I. INTRODUCTION

The fast wireless communications has always been accompanied by an equally significant increase in the daily data creation. Thus, modern communication systems are always at the peak of their performance, reflecting the usage of the available spectrum, and at the same time keeping within the limits of the prescribed energy consumption. Existing communication systems transmit all incoming data, regardless

of whether the information is relevant to the purpose or not. This, in turn, leads to excessive and unwarranted usage of bandwidth and power, which is avoidable [1].

To mitigate the negative consequences, semantic communication has started to be explored deeper and deeper by the researchers as the most promising solution. Unlike normal communication, which implies transferring all the bits, semantic communication only selects and imparts the information that has meaning for the receiver. It in fact, translates the raw symbols to the underlying intent, a concept in Shannon's original communication theory [2] but it also goes beyond syntactic correctness and relies on the transmitted content's relevance and usefulness. This way communication systems become more effective and are able to cope with the increasing data volume of today [1].

The benefits are even clearer when the two modes of communication – that of semantic and D2D - are combined. The closest device data sharing reduces the need for the entire traffic of communication to pass through the base station. The network is made more reactive and resource-efficient through the shorter transmission path that cuts latency and energy consumption [3].

This study investigates the connection between energy efficiency (EE) and spectral efficiency (SE) in a D2D-based framework that incorporates semantic communication. The system was assessed under varying SNR conditions, and a non-linear trend was noticed: initially, EE goes up with rising SNR and SE, but after a certain point, improvement gets slow and eventually stops. This defines a real limit beyond which more energy is required to attain better performance. The main idea is that the combination of D2D communication with semantic processing results in a change in network design philosophy - from merely maximizing performance to default networks that

are efficient and sustainable [4].

A. Related Work and Technological Evolution

The progression of technology from Shannon communication to semantic networks represents a conceptual change. Shannon's mathematical theory of communication [2] established the fundamental limits of reliable communication over noisy channels, dealing mainly with the syntactic correctness at the bit level. Luo et al. suggest that this entails a transition from syntactic fidelity to semantics, where efficiency is not based on symbols but rather on meaning [5]. Following this path, Kutay Andener introduced dynamic game-based theoretical frameworks and semantic distortion metrics, thus paving the way for intelligent task-oriented communication systems [3]. Four evolutionary stages of semantic communication were identified by Liu et al. [1]:

- Shannon-level bit transmission.
- Compression and representation at the semantic level.
- Semantic communication occurs in several forms(text, image, audio).
- 6G and the advancement of AI offer intelligent cooperative communication [6].

B. Research Objectives and Key Contributions

The research intends to assess and measure the compromise in communication of the spectral efficiency (SE) and energy efficiency (EE) with the collaboration of device-to-device (D2D) in semantic communication being the main motor of this trade-off. This study takes the generalized understanding of semantic communication from prior studies, together with the quantitative simulation results, as affirmation of the potential advantages where semantic communication is involved [1].

Specific Objectives

- 1) **To simulate the effects of semantic compression on spectral and energy efficiencies.**

Using mathematical computations, SE and EE are analyzed at their respective levels of signal-to-noise ratio (SNR) with semantic compression ratios (0.1-0.7) applied to the modulation schemes that are conventional [4].

- 2) **To incorporate D2D cooperation gains into semantic communication frameworks.**

In the cooperative model, the robustness, power savings, and information fidelity achieved through the joint semantic and D2D communication are analyzed. [3] Also, the resource allocation strategies for D2D underlay networks are put under scrutiny to ensure that system performance is at its best [7].

Key Contributions

- **Hybrid Model Development:** A new framework that combines Shannon's capacity [2], cooperation gains, and semantic encoding for the simulation of semantic-D2D communication has been developed [5].

- **Energy-Spectral Tradeoff Characterization:** The statement of meaning-based transfer has been extended by quantitative evidences that made EE better with a minor rise in SE cost [8].
- **Knowledge Base Alignment Conceptualization:** The approach includes semantic inference and D2D cooperation in a common body of knowledge and is based on theoretical perspectives of earlier research. [1].

C. Real-World Impact and Applications

The integration of semantic communication and device-to-device (D2D) systems will be a key factor in the operation of the coming 6G technologies in the real-world [6].

- **Smart IoT and Edge Computing:** Rather than sending every single unit of data, IoT devices will have the capacity to focus on what is really important. The application of semantic compression will make them communicate only the important matter, thus the bandwidth and energy will be saved. The spectrum allocation based on the semantic awareness will promote the efficient utilization of the spectrum to be allocated among the IoT networks [4], [9].
- **Autonomous Vehicles:** The D2D-based semantic communication will enable the vehicle-to-vehicle interaction which is the low-latency and meaning-aware (e.g. "pedestrian ahead" vs. "object detected") exchanges [3].
- **Healthcare and Remote Sensing:** Through the process of semantic data fusion, one can achieve a reliable diagnosis and monitoring, even during bad channel conditions [1].
- **Digital Twins and Metaverse Systems:** Meaning-preserving communication guarantees the synchronization of the digital and physical worlds. Therefore, the virtual models are updated in real-time continuously, mirroring the physical changes without any latency [6].

II. LITERATURE REVIEW

A. Development of Semantic Communication

When we consider the work of communication research, it largely remained in the concept of Shannon of 1948: transmit bits in the right way and deliver more of them over the channel [2]. Long have we been contented with that. It made engineers work out the way to eliminate noise and achieve the maximum of what the channel is capable of. However, today, with the advent of such systems as 6G, digital twins, and the metaverse, it is becoming apparent that it is no longer the quantity of bits that matters when it comes to the story [1]. Humans and machines are not just interested in perfect symbols, they desire the content of them. The reason why semantic communication (SC) emerges is that in actual conversations we are concerned with being understood. Luo et al. and Kutay & Yener go further to add that SC combines wireless networks with AI and linguistics in such a way that the receivers can still get the intended meaning, despite the signals not being perfect [5]. It cuts out extraneous information, and, strangely enough, attacks the ancient boundaries Shannon had provided us.

B. Significant Technology in the Semantic Communication

1) *Semantic Extraction and Representation*: In a modern algorithm, particularly in NLP, computer vision, attempts to extract the important meaning. Similar to what Liu et al. indicate, semantic encoders are capable of reducing this to small vectors such that the system does not expend bandwidth on items that no one requires [1].

2) *Knowledge Base and Contextual Inference*: The other issue that Luo et al. emphasize is that there is a certain background knowledge that the sender and receiver share that helps the receiver to infer the missing parts in case something is lost [5]. Humans already do this, right? When one forgets a word, we know the point anyway.

3) *Joint Source Channel Semantic Coding*: Compression and channel coding are combined in a single shot in networks like DeepSC, IS-JSCC and SGSCC. This normally prevents any interference and minimizes the amount of power consumption simultaneously [6].

C. Coexistence with Device-to-Device (D2D) Communication

In the SC case where D2D is taking place, local users are able to converse directly to one another, as far as is concerned locally. One of the most pleasant side effects is that energy and even semantic information (such as context) can be shared across devices. Previous D2D research on the interference and efficiency assists in explaining the handling of the same well [7] Provided that there is a low level of interference and devices possess sufficient power, D2D links are able to eke extra out of the accessible bandwidth. Teamwork enhances semantic throughput at that time and reduces energy consumption since the nodes will no longer be operating independently [3].

D. Metrics and Tradeoffs of Performance

SC is concerned about other objectives. Rather than just discussing BER or spectral efficiency, scientists discuss semantic accuracy and the extent of confusion of the meaning [2]. Kutay and Yener even do not refer to the common BER at all but examine the semantic distortion with the purpose of protection of the sent idea. In principle and practice, spectral efficiency reaches a limit as the cooperation becomes limited. However, the efficiency of energy tends to increase with the efficiency of compression - more data compression means less power. SC with D2D will now have a nonlinear energyspectral efficiency trade that happens to be the destination of the modern networks anyway [8].

III. SIMULATION SETUP AND DATA GENERATION

The performance of semantic communication systems incorporated with device-to-device (D2D) underlay networks was comprehensively evaluated by the study through a simulation-based method. A unified analytics platform based on Python was used to synthetically produce all the quantitative performance metrics consisting of Energy Efficiency (EE), Spectral Efficiency (SE), Quality (Q), and D2D performance features. Located on analytically modeled path-loss channel conditions,

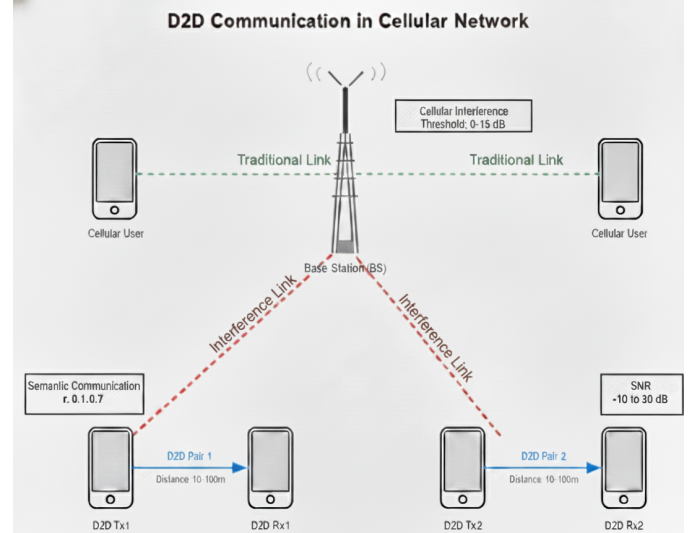


Fig. 1. Representation of D2D communication in a cellular network.

the simulation environment includes semantic compression modeling, cooperative D2D communication, and energy spectral tradeoff evaluation [4].

A. Simulation Parameters

The simulation setup was adjusted according to the parameters listed in Table I, which represented realistic wireless communication situations for urban D2D networks inferred from widely accepted channel models [10]. The path loss and propagation properties are in line with the standard analytical models for D2D underlay cellular networks [7].

TABLE I
SIMULATION PARAMETERS

Parameter	Value(s)
Modulation Schemes	QPSK ($M = 4$), 16-QAM ($M = 16$)
Semantic Compression Ratio (r)	{0.1, 0.3, 0.5, 0.7}
Coding Rate (R_c)	0.8
D2D Distance	10–100 m (step: 10 m)
Cellular Interference Threshold	{0, 5, 10, 15} dB
SNR Range	–10 to 30 dB (step: 2 dB)
Transmit Power	20 dBm

The system examines different semantic compression ratios $r = \{0.1, 0.3, 0.5, 0.7\}$. higher values retain more contextual information. The old method of modulation utilized QPSK with $M = 4$ and semantic signaling exercised 16-QAM with $M = 16$. The distances for D2D underlay range from 10 m to 100 m while the cellular interference limits are set at 0, 5, 10, and 15 dB. The path loss exponent of 3.5 is used to simulate the urban propagation environment. This arrangement is exactly similar to Python configuration towards result generation [10].

B. Synthetic Data Generation

All results are obtained from Python-based simulation, where SNR, path loss, SINR, SE, EE, and quality values are computed analytically to form the synthetic dataset.

1) Channel and SNR Generation

The wireless channel follows a simple log-distance path-loss model:

$$PL(d) = 40 + 35 \log_{10}(d), \quad (1)$$

where d is the transmitter–receiver distance in meters.

SNR values are generated in the range:

$$SNR_{dB} \in [-10, 30], \quad (2)$$

and converted to linear scale using:

$$SNR = 10^{SNR_{dB}/10}. \quad (3)$$

2) Semantic Compression Levels

A natural-language message is compressed at four levels:

$$r \in \{0.1, 0.3, 0.5, 0.7\}.$$

For each level, a shorter message is created. The effective semantic retention is:

$$r_{\text{eff}} = \frac{\text{compressed word count}}{\text{original word count}}. \quad (4)$$

3) Traditional Communication Metrics

Traditional QPSK transmission uses the following models: Spectral efficiency:

$$SE_{\text{trad}} = \log_2(4) \log_2(1 + SNR), \quad (5)$$

Power consumption:

$$P_{\text{trad}} = \frac{1}{1 + SNR}, \quad (6)$$

Energy efficiency:

$$EE_{\text{trad}} = \frac{SE_{\text{trad}}}{P_{\text{trad}}}. \quad (7)$$

A smooth saturation function is used to represent quality:

$$Q_{\text{trad}} = \tanh(0.15(SNR_{dB} + 10)).$$

4) Semantic Communication Metrics

Semantic communication uses 16-QAM with coding rate $R_c = 0.8$.

Spectral efficiency:

$$SE_{\text{sem}} = \log_2(16) R_c \log_2(1 + SNR) r_{\text{eff}}. \quad (8)$$

Power model:

$$P_{\text{sem}} = \frac{1}{1 + SNR} (1 - 0.5r_{\text{eff}}). \quad (9)$$

Energy efficiency:

$$EE_{\text{sem}} = \frac{SE_{\text{sem}}}{P_{\text{sem}}}.$$

Quality metric:

$$Q_{\text{sem}} = \tanh(0.18(SNR_{dB} + 10)r_{\text{eff}}).$$

5) Cooperative Semantic D2D Model

A simple cooperation gain is applied as:

$$G_{\text{coop}} = 1 + 0.4r_{\text{eff}}.$$

6) D2D Underlay SINR

For D2D underlay communication, SINR is computed as:

$$SINR = \frac{P_{\text{rx}}}{P_{\text{noise}} + P_{\text{int}}}. \quad (10)$$

Distances from 10–100 m and different interference thresholds are used to generate SE, EE, and SINR curves for underlay D2D links.

All plots in this work are produced using this synthetic dataset.

C. Visualization and Analysis

The performance analysis comprises the following six evaluations:

- 1) The comparison of traditional, semantic, and semantic-D2D models with respect to energy efficiency vs SNR;
- 2) All compression ratios will be compared under the criteria of Spectral Efficiency vs SNR;
- 3) Semantic Quality vs SNR screening for the recovery of perceived meanings.
- 4) The tradeoff between Energy Efficiency and Spectral Efficiency (EE-SE) mapping.
- 5) Energy Efficiency vs D2D Distance for different thresholds of cellular interference.
- 6) Comparison of different interference levels with respect to Spectral Efficiency vs D2D Distance.

IV. SEMANTIC - D2D SYSTEM ARCHITECTURE ANALYSIS

In general, the system architecture consists of 3 levels of hierarchy: (i) standard modulation; (ii) semantic representation and its compression; (iii) D2D (device-to-device) cooperation. This stacking of layers brings advantages in both semantic efficiency and local cooperation, influencing the gains in power and spectral performance. The synchronization between design and simulation model demonstrates the coherence of conceptual modeling with implementation [4].

A. Traditional Baseline Communication Layer

The traditional communication baseline transmits whole symbol-level data by means of QPSK ($M = 4$) modulation. This focus on bits defines the benchmark for assessing the benefits brought about by semantic compression and cooperation. So, even though the performance is the same throughout the SNR range, it is less energy-efficient since unnecessary bits are transmitted without taking the context into account. The baseline adheres to Shannon's classical communication theory [2] [10].

B. Semantic Representation Layer

The layer of semantic communication works by selectively passing only the significant content, using the compression ratios $r = \{0.1, 0.3, 0.5, 0.7\}$. The current research utilizes semantic compression by r scaling of spectral efficiency and transmission power analytically, instead of whole text-based semantic encoding. The 16-QAM ($M = 16$) modulation scheme has been employed, which is superior in terms of symbol capacity compared to QPSK baseline. The higher r values retain more information, whereas the smaller r values save more power since fewer symbols are transmitted. Through meaning-aware optimization, this layer automatically and concurrently increases Spectral Efficiency (SE) and Energy Efficiency (EE) [6].

C. Hybrid Semantic - D2D Integration

The integration of semantic and D2D layers results in a hybrid communication system. Global semantic modeling is an approach that leads to reduction in the number of symbol transmissions, while D2D collaboration brings local clarity and enhances the meaning of symbols, even if the SNR or interference is low. The cooperative gain, $G_{\text{coop}} = 1 + 0.4r$, which has been modeled in the simulation, gives a quantitative expression of this increase. This enables the system to make a dynamic adjustment of the allocation among bandwidth, quality of meaning, and energy savings. It is an intelligent spectrum management network that is connected through [9] semantic-aware resource allocation strategies that are enabled. The Semantic-D2D configuration, in line with the trends observed in the simulations, achieves the best Energy Efficiency and Spectral Efficiency blend at heavier compression ratios ($r \geq 0.5$) [4].

V. RESULTS

A. Performance Evaluation Based on SNR

1) Energy Efficiency vs SNR

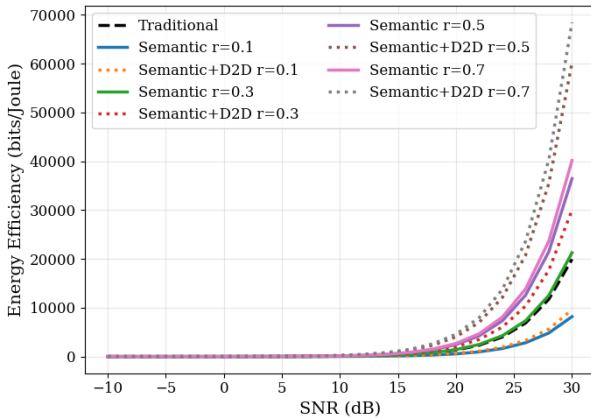


Fig. 2. Energy Efficiency vs SNR for traditional, semantic, and Semantic-D2D communication models under various compression ratios.

Figure 2 The analysis reveals that Energy Efficiency (EE) exhibits exponential growth with increasing Signal-to-Noise

Ratio (SNR) across all evaluated communication schemes. Our results show that the semantic compression strategy, especially when used together with the Device-to-Device (D2D) cooperative transmission, provides significantly greater energy efficiency as compared to traditional ones. This is a performance improvement that is more pronounced at high compression ratios. This is due to two main reasons: the lower energy consumption of required data transmission and intelligent choice of information that can be sent when it is semantically relevant, and not whole raw data. The system is more resource utilizing because it attends to a meaning, as opposed to bits. The trends observed are in line with the underlying energy-spectral efficiency trade-offs reported in wireless communication literature, and show that semantic-aware techniques can positively offset this trade-off line [8] [4].

2) Spectral Efficiency vs SNR

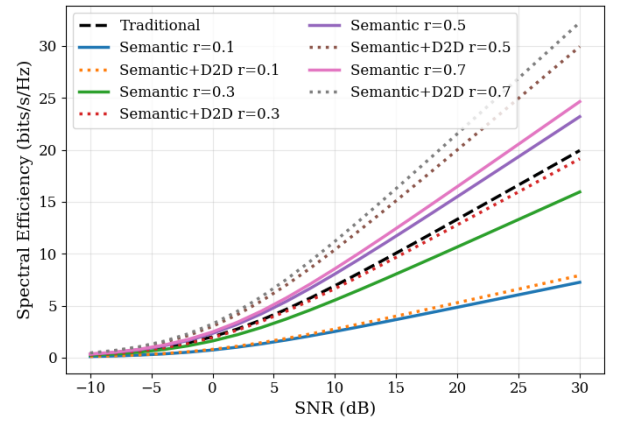


Fig. 3. Spectral Efficiency vs SNR comparison among traditional, semantic, and Semantic-D2D schemes for different compression ratios.

Figure 3 The spectral efficiency (SE) analysis shows that the semantic communication is always superior to the traditional systems in terms of performance at different SNRs because representation of the symbols is optimized. SE gains through spatial diversity are added by the semantic-D2D cooperative scheme. An increase in the compression ratio makes this performance improvement significant, and semantic methods are more effective in preserving information quality.

Additionally, the findings show that the advantages of semantic encoding are proportional to the quality of channels. The compressed semantic symbols at moderate and high SNRs have more contextual meaning per transmitted bit, leading to excellent spectral utilization than the baseline model. The cooperating D2D scheme also reduces the burden on the cellular link by taking advantage of close-range relaying thus allowing a more efficient reuse of the spectrum. As a result, it also implies that even with attenuation in the wireless channel, semantic and semantic-aided D2D communication still maintains a higher level of throughput compared to traditional transmission schemes that use bits [6].

3) Quality Metric vs SNR

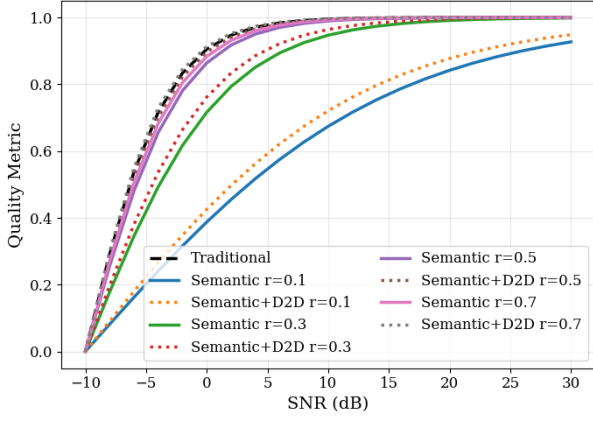


Fig. 4. Quality metric performance versus SNR for traditional, semantic, and Semantic-D2D communication frameworks.

Figure 4 shows the correlation between Quality Metric and SNR. The findings indicate that semantic and semantic-D2D systems have a greater perceived quality despite the lower SNR. The quality measure becomes saturated with increased SNR implying that the semantic layer does not rely on the performance of the physical layer, and it is more context-sensitive. This action emphasizes the strength of semantic representation where the symbols that are transmitted are aimed at transferring vital meaning as opposed to a full reconstruction at bit level [6].

4) Energy Efficiency vs Spectral Efficiency Tradeoff

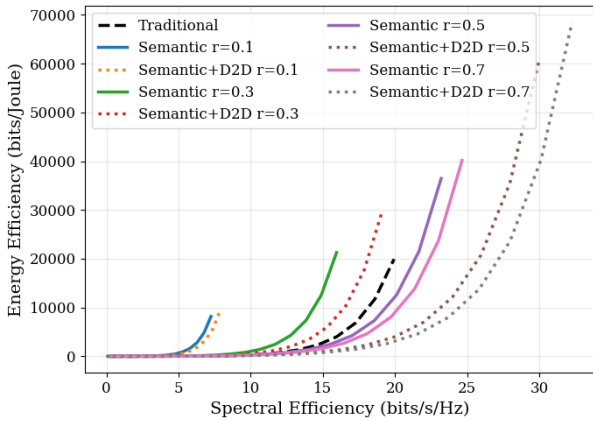


Fig. 5. Energy Efficiency vs Spectral Efficiency tradeoff for traditional, semantic, and Semantic-D2D communication models under different compression ratios.

Figure 5 shows the tradeoff correlation between Energy Efficiency (EE) and Spectral Efficiency (SE). The EE-SE balance of Semantic and Semantic-D2D systems is higher than that of the traditional one. In the condition of higher compression ratios, Semantic-D2D is more effective, as cooperative gains and effective semantic-level compression allow to optimize the

use of energy and bandwidth. This is a characteristic of energy-spectral efficiency tradeoffs in current wireless structures [8].

B. Energy Efficiency vs D2D Distance

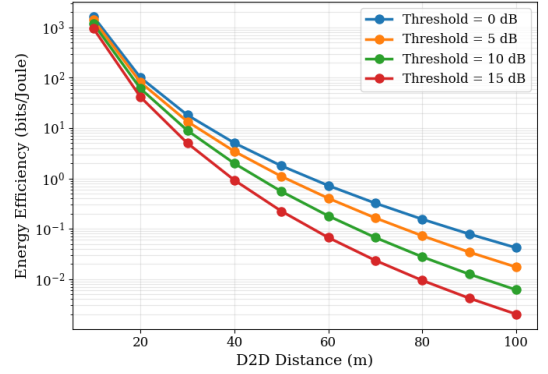


Fig. 6. Energy Efficiency versus D2D distance for different cellular interference thresholds at compression ratio $r = 0.5$.

Figure 6 presents the variation of Energy Efficiency (EE) with respect to D2D distance for a fixed compression ratio $r = 0.5$ under different cellular interference thresholds. Due to increased path loss and decreased received signal power, the results show that EE decreases exponentially with increasing D2D distance. Higher thresholds lead to degradation because of stronger cellular interference, whereas lower thresholds (e.g., 0 dB) produce better EE performance [7]. Overall, the trend shows that in order to fully benefit from semantic compression, optimal D2D operation necessitates carefully balancing interference and distance constraints.

C. Spectral Efficiency vs D2D Distance

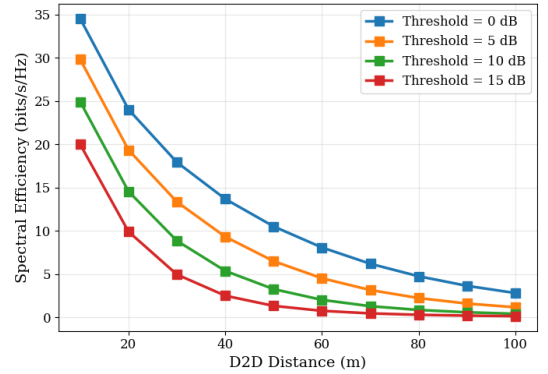


Fig. 7. Spectral Efficiency versus D2D distance for different cellular interference thresholds at compression ratio $r = 0.5$.

Figure 7 demonstrates how, at different interference thresholds, the Spectral Efficiency (SE) varies with D2D distance for the same compression ratio $r = 0.5$. Path loss and interference effects cause SE to gradually decrease as the D2D distance increases. Semantic-D2D communication achieves higher SE values at lower interference levels, demonstrating its

efficacy in preserving dependable data exchange over moderate distances with the advantages of semantic compression [4].

VI. CONCLUSION

The trade-off between Energy Efficiency (EE) and Spectral Efficiency (SE) in semantic communication systems improved by Device-to-Device (D2D) cooperation was thoroughly examined in this paper. To model QPSK and 16-QAM systems under various SNR conditions, semantic compression ratios, and D2D distances, a comprehensive simulation framework based on Python was created. The findings showed that while D2D cooperation further improves both EE and SE through cooperative gain and short-range communication, semantic compression dramatically improves energy utilization by reducing redundant symbol transmissions.

Semantic-D2D configuration was found to result in the maximum energy efficiency (EE) in all the simulated conditions, especially in medium and high SNR conditions. This is due to the adaptive nature of semantic compression, effective bandwidth reuse, and reduced path loss. The results verify that collaborative D2D communication using meaning-aware transmission provides a sustainable and efficient basis for next-generation wireless networks.

Although the current study involves the use of an analytical simulation model, a number of extensions are to be made in future work:

- **Adaptive Semantic Compression:** Future systems can optimize performance even further by dynamically changing the compression ratio r in response to channel conditions or instantaneous SNR.
- **Realistic Channel Modeling:** Shadowing, fading, and non-line-of-sight (NLOS) propagation effects can be included to enhance the accuracy of channel prediction.
- **Learning-Based Semantic Encoding:** Deep neural encoders provide an opportunity to learn semantic features automatically instead of relying on analytical scaling, thereby preserving meaning across multiple modalities such as text, audio, and images.
- **Multi-User and Multi-Hop Scenarios:** The model can be further generalized to cooperative relay systems and multi-user D2D setups in order to investigate scalability and spectral-sharing performance.

Overall, this work provides a clear path for experimental and AI-enhanced extensions in 6G semantic networking and validates a foundation for meaning-driven, energy-efficient wireless communication.

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